CNOT Circuits and Phase-free ZX Diagrams

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Schedule

- This week: CNOT universality
- Next week: CNOT completeness

The field $\mathbb{F}_2$

Definition 4.1.1.

$$\mathbb{F}_2 = \{0, 1\}, \qquad a + b = a \oplus b \text{ (XOR)}, \qquad ab = a \wedge b \text{ (AND)}.$$

1. **Associativity:** $(a + b) + c = a + (b + c)$; $(ab)c = a(bc)$.
2. **Commutativity:** $a + b = b + a$; $ab = ba$.
3. **Identity:** $a + 0 = a$; $a \cdot 1 = a$; $0 \neq 1$.
4. **Inverses:** $a + a = 0$; $a \neq 0 \Rightarrow a = 1 = a^{-1}$.
5. **Distributivity:** $a(b + c) = ab + ac$.

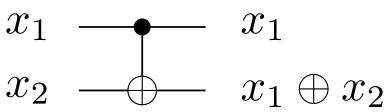
$$\mathbb{F}_2^n = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} : x_i \in \mathbb{F}_2 \right\}$$

bit string

CNOT

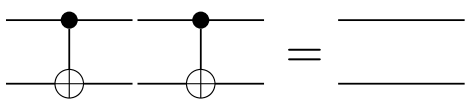
Definition *1.

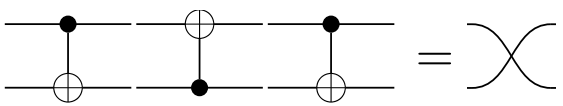
$\mathbf{CNOT}$ is the PROP of circuits generated by CNOT, identity wires, and wire permutations.

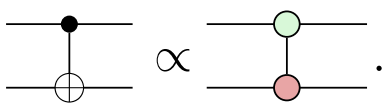


Write c_{ij} for control i, target j.

Identities. CNOT is self-adjoint and unitary. Its circuit identities and ZX representation are:







Parity Maps and Matrices

Definition 4.1.2. Parity map

A map $f : \mathbb{F}_2^n \to \mathbb{F}_2^m$ is a *parity map* if each output is the sum in $\mathbb{F}_2$ of a subset of the input bits.

Definition 4.1.4. Parity matrix

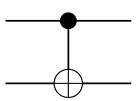
A *parity matrix* has entries in $\mathbb{F}_2$. The matrix of f is given by $A_{ji} = 1$ iff output j includes input i. Hence $f(x) = Ax$.

Example. Parity map

$$f : \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \longmapsto \begin{pmatrix} x_1 + x_3 \\ x_2 + x_3 \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}}_{A} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 + x_3 \\ x_2 + x_3 \end{pmatrix}$$

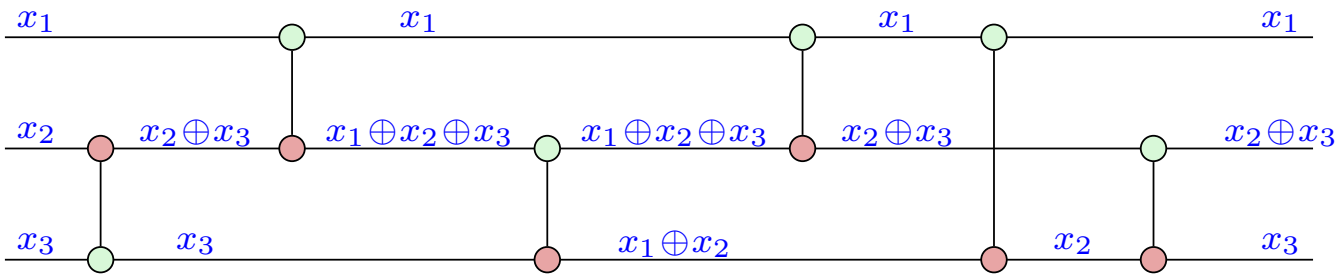
Example. CNOT



$$: \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \longmapsto \begin{pmatrix} x_1 \\ x_1 \oplus x_2 \end{pmatrix}$$

$$\text{so } \underbrace{\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}}_{A} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_1 + x_2 \end{pmatrix}$$

CNOT parity maps



(4.1)

$$U \;::\; |x_1, x_2, x_3\rangle \mapsto |x_1, x_2 \oplus x_3, x_3\rangle$$

Proposition 4.1.3.

Every CNOT circuit C on n wires implements a parity map $x \mapsto Ax$ for an invertible parity matrix $A \in \mathbb{F}_2^{n \times n}$.

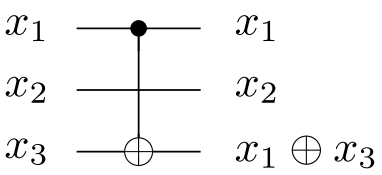
Elementary Parity Matrices

Definition *6.

For $i \neq j$, the elementary parity matrix E_{ij} is the $n \times n$ identity matrix with an additional 1 in row j, column i.

It adds bit i to bit j and leaves every other bit unchanged.

Example. Three wires, control 1 and target 3.



$$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}}_{E_{13}} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_1 + x_3 \end{pmatrix}$$

CNOT synthesis

Theorem 4.1.7.

For every invertible $n \times n$ parity matrix A, Algorithm 1 returns a CNOT circuit implementing A.

Algorithm 1. CNOT synthesis by Gaussian elimination

Given an invertible $n \times n$ parity matrix A, record these row additions:

1. For $i = 1, \ldots, n$, if $A_{ii} = 0$, add a row $k > i$ with $A_{ki} = 1$ to row i. Then add row i to every row $j > i$ with $A_{ji} = 1$.
2. For $i = n, \ldots, 1$, add row i to every row $j < i$ with $A_{ji} = 1$.
3. Return the corresponding CNOT gates in reverse order.

https://www.statlect.com/matrix-algebra/Gaussian-elimination

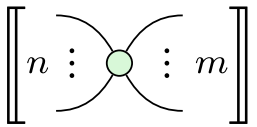
Same as usual gaussian elimination, but:

- In $\mathbb{F}_2$, addition is XOR and nonzero pivots are 1, so we do not rescale rows.
- If a pivot is 0, we add a row instead of swapping rows.
- We clear above and below the diagonal to get I (Gauss–Jordan elimination).
- We read off the CNOTs in reverse order to get a circuit for A.

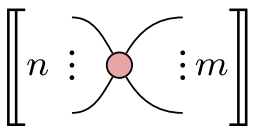
Phase-free ZX

Generators and interpretation

A phase-free ZX diagram is generated by the following spiders and structural maps.



$$= |0\rangle^{\otimes m}\langle 0|^{\otimes n} + |1\rangle^{\otimes m}\langle 1|^{\otimes n}$$



$$= |+\rangle^{\otimes m}\langle +|^{\otimes n} + |-\rangle^{\otimes m}\langle -|^{\otimes n}$$

$$\left(|0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}, \quad |\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}\right)$$

$$= \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$$

$$= \begin{pmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 1\end{pmatrix}$$



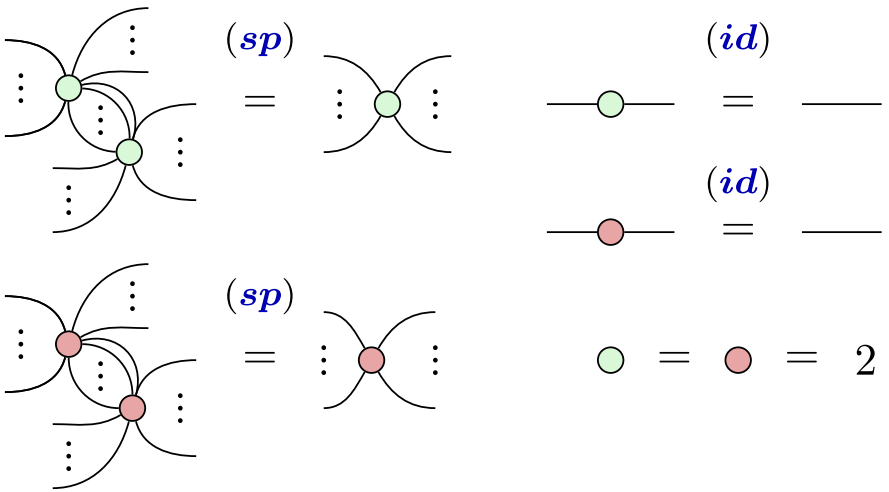
$$= |00\rangle + |11\rangle$$

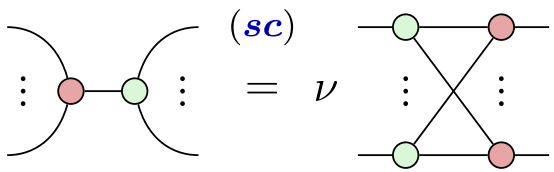


$$= \langle 00| + \langle 11|$$

Phase-free ZX

Wire bending and the following equations define the calculus.



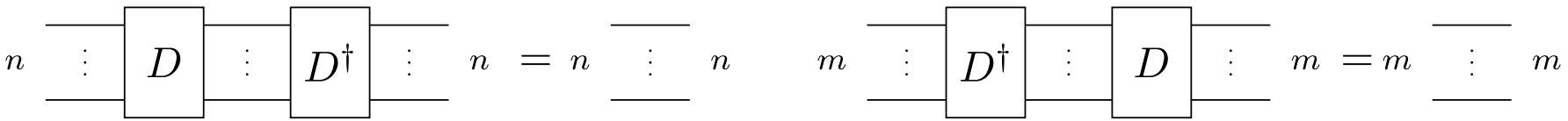


In (sc), $\nu = 2^{(n-1)(m-1)/2}$. Throughout, nonzero overall scalars are ignored, as in the chapter.

Dagger and unitarity

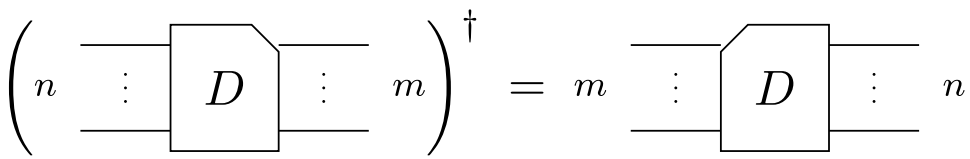
Definition *3. Unitarity

A diagram $D : n \to m$ is called unitary if the following holds:



Definition *7. Dagger

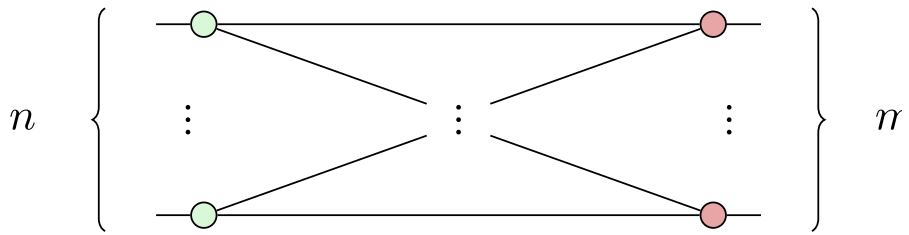
Given a diagram $D : n \to m$, we define its dagger $D^\dagger : m \to n$ by flipping it horizontally:



Parity normal form

Definition 4.2.2.

A *PNF* has one Z-spider per input and one X-spider per output, each with exactly one boundary wire. All other wires join opposite colours, at most once per pair. There are no other spiders or components.

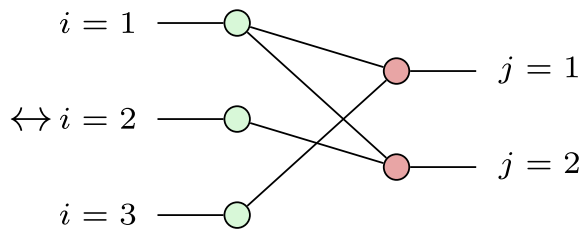


Biadjacency matrix

Definition *4.

We define the biadjacency matrix $A = (A_{ji})$ by $A_{ji} = 1$ iff input Z-spider i is connected to output X-spider j. Write $P(A)$ for this diagram. Columns are inputs and rows are outputs.

$$\begin{array}{c} \\ j \downarrow \\ \\ \end{array} \begin{array}{c} \longrightarrow i \\ \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \end{array} \begin{array}{l} j=1 \\ j=2 \end{array}$$

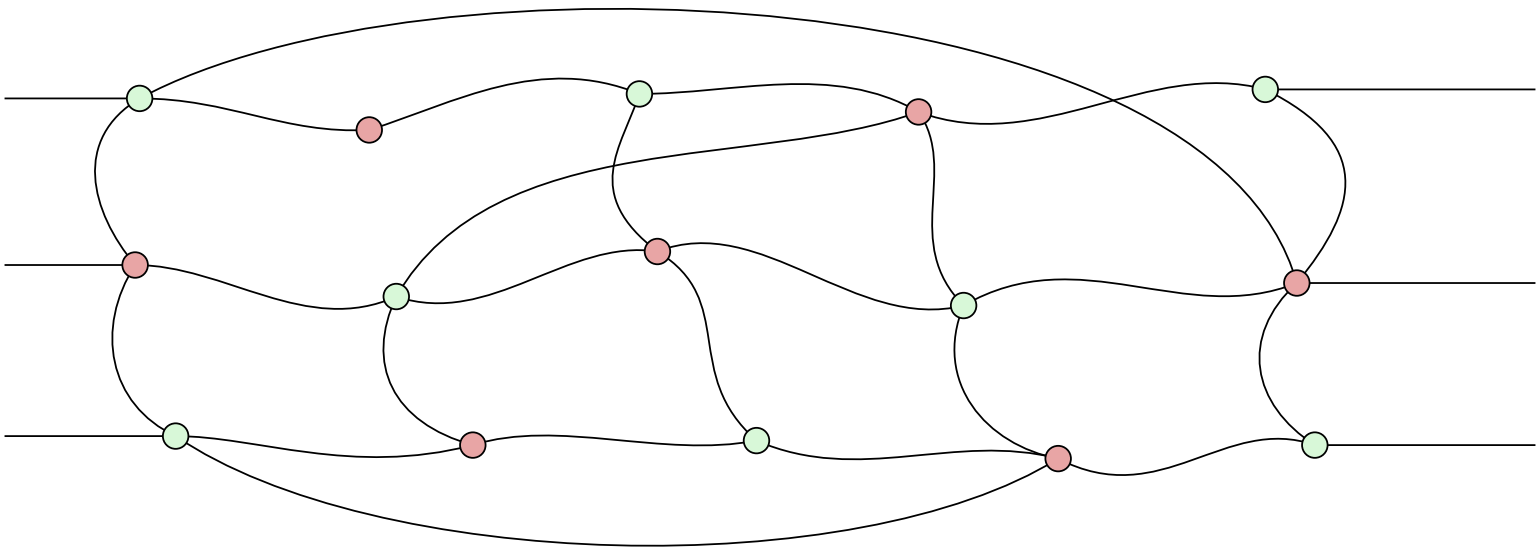


Two-coloured form

Definition 4.2.4.

A diagram is *two-coloured* if no two spiders of the same colour are connected, and there are no self-loops, parallel wires or Hadamards.

Example.



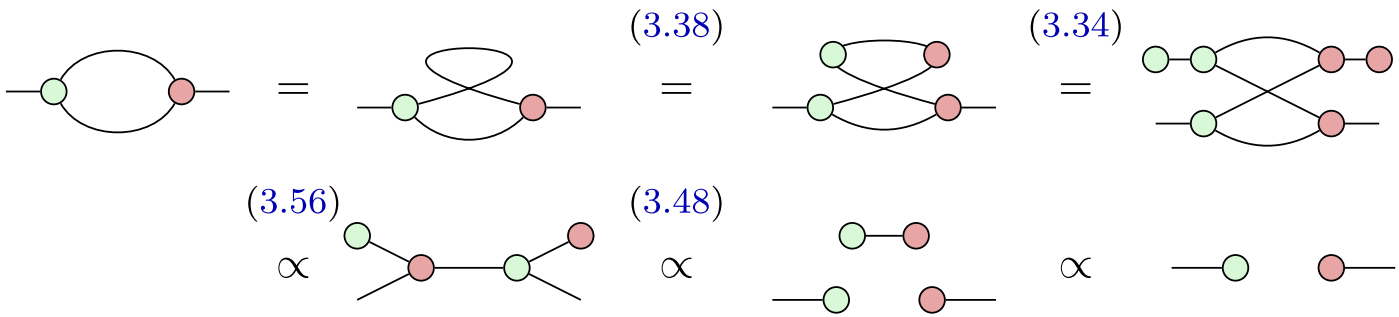
Two-coloured form

Lemma *4.2.5.

Every phase-free ZX diagram can be efficiently simplified to a two-coloured one.

Proof.

1. Fuse connected spiders of the same colour.
2. Insert an identity spider on each self-loop and fuse it away.
3. Finally, remove pairs of parallel wires by the Hopf rule:



Their endpoints have different colours, so the rule applies. Each step decreases the number of spiders or wires.

Input, output and internal spiders

Definition 4.2.6.

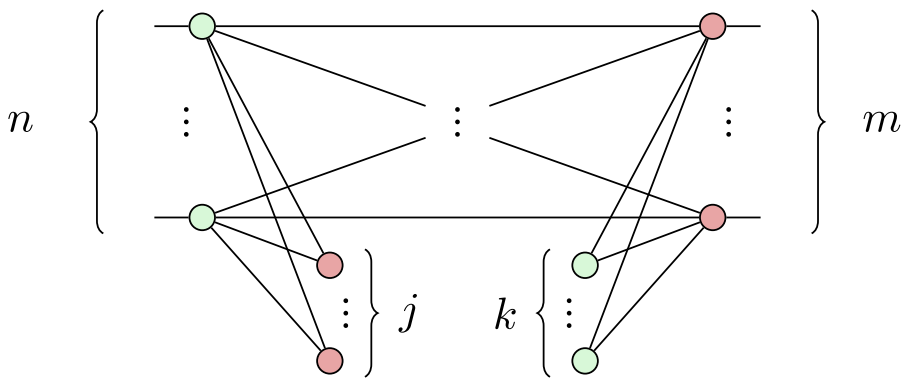
A spider is an *input* or *output spider* if connected to an input or output, respectively. It is *internal* if it is neither.

Generalised parity form

Definition 4.2.7.

A two-coloured diagram is in *GPF* when:

1. every input meets a Z-spider and every output meets an X-spider;
2. each spider meets at most one input or output;
3. there are no zero-arity spiders;
4. no two internal spiders are connected.



Here j counts internal X-spiders and k counts internal Z-spiders. PNF has $j = k = 0$.

Algorithm 2: Reducing to generalised parity form

Start with a two-coloured phase-free diagram.

1. Apply spider fusion and the cleanup in Lemma 4.2.5. Remove zero-arity spiders and bare closed wires, recording a factor of 2 for each.
2. If a Z-spider that is not an input is connected to an X-spider that is not an output, apply (sc).
3. If step 2 applied, return to step 1. Otherwise go to step 4.
4. Use (id) in reverse so that every input has its own Z-spider and every output its own X-spider.

Generalised parity form: termination

Algorithm 2.

Start with a two-coloured phase-free diagram.

1. Apply spider fusion and the cleanup in Lemma 4.2.5. Remove zero-arity spiders and bare closed wires, recording a factor of 2 for each.
2. If a Z-spider that is not an input is connected to an X-spider that is not an output, apply (sc).
3. If step 2 applied, return to step 1. Otherwise go to step 4.
4. Use (id) in reverse so that every input has its own Z-spider and every output its own X-spider.

Lemma 4.2.8.

Algorithm 2 terminates efficiently with a diagram in GPF.

Proof.

- Step 2 removes a non-input Z-spider.
- Each new Z-spider either becomes an input or disappears by fusion in step 1.
- Thus each iteration decreases the number of non-input Z-spiders.
- New X-spiders similarly become outputs or fuse away, so only boundary spiders survive as new vertices.
- There are at most as many iterations as original Z-spiders, each with polynomial cost.

Generalised parity form: boundary repair

Algorithm 2.

Start with a two-coloured phase-free diagram.

1. Apply spider fusion and the cleanup in Lemma 4.2.5. Remove zero-arity spiders and bare closed wires, recording a factor of 2 for each.
2. If a Z-spider that is not an input is connected to an X-spider that is not an output, apply (sc).
3. If step 2 applied, return to step 1. Otherwise go to step 4.
4. Use (id) in reverse so that every input has its own Z-spider and every output its own X-spider.

Lemma 4.2.8.

Algorithm 2 terminates efficiently with a diagram in GPF.

Proof (continued).

- When step 2 no longer applies, no internal spiders are connected.
- For step 4, retain one correct boundary attachment per old spider.
- On each other attachment insert one identity if its colour is wrong, two if it is correct.
- On bare wires use the shortest alternating chain with the required endpoint colours.
- New internal spiders meet only boundary spiders, so conditions 1–2 hold without affecting 3–4.

Unitarity forces parity normal form

Proposition 4.2.9.

If a GPF diagram is proportional to an isometry, then $j = 0$; if proportional to a unitary, then $j = k = 0$. In particular, a unitary in GPF is already in PNF.

Proof.

Suppose an internal X-spider exists. Its neighbours are input Z-spiders. Put $|1\rangle$ on one of these inputs and $|0\rangle$ on all other inputs. The Z-spiders copy the bits, so this internal spider evaluates to

$$\left(\langle +|^{\otimes d} + \langle -|^{\otimes d}\right)|10\cdots 0\rangle = 2^{-d/2}(1-1) = 0.$$

Thus a nonzero input is sent to zero, contradicting isometry. Hence $j = 0$.

Similarly, an internal Z-spider has only output X-spiders as neighbours. Put $\langle -|$ on one of these outputs and $\langle +|$ on all others. Copying these effects backwards gives

$$\left(\langle -| \otimes \langle +|^{\otimes (d-1)}\right)\left(|0\rangle^{\otimes d} + |1\rangle^{\otimes d}\right) = 2^{-d/2}(1-1) = 0.$$

This gives a nonzero covector η with $\eta D = 0$, contradicting invertibility. Hence $k = 0$.

■

Parity normal form

Theorem 4.2.10.

Any unitary phase-free ZX diagram can be efficiently rewritten in PNF.

Proof.

First use Lemma 4.2.5 to reduce it to two-coloured form. Then apply Algorithm 2 to reduce it to GPF. It is proportional to a unitary, so Proposition 4.2.9 makes it PNF.

■

Unitarity and invertibility

Lemma *4.5.

A PNF is proportional to a unitary iff its biadjacency matrix A is square and invertible.

Proof.

On a basis input $|x\rangle$, Z-spiders copy the bits and X-spiders compute their parities. The scalar factors depend only on the numbers of wires, so

$$[\![P(A)]\!]|x\rangle = c_A|Ax\rangle, \qquad c_A \neq 0.$$

If A is invertible, this permutes the basis. Conversely, unitarity forces equal dimensions. If A were singular, choose $z \neq 0$ with $Az = 0$; then $[\![P(A)]\!](|0\rangle - |z\rangle) = 0$, contradicting injectivity.

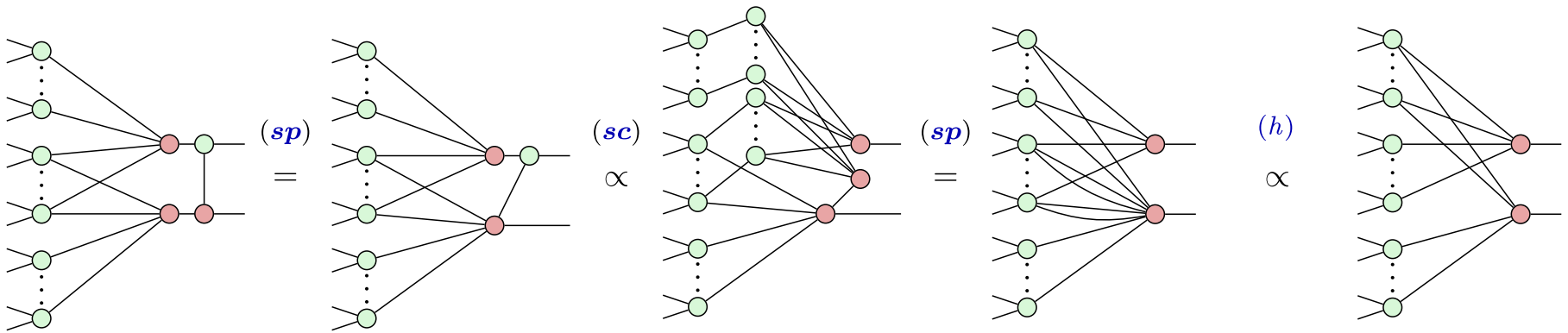
■

CNOT row addition

Lemma 4.2.11.

Attaching a CNOT with control i, target j to a PNF changes its target row by $r_j \leftarrow r_j + r_i$.

Proof.



Fusion, strong complementarity, fusion, then Hopf cancel shared connections. The target keeps exactly those present in one row only: row addition over $\mathbb{F}_2$.

CNOT extraction

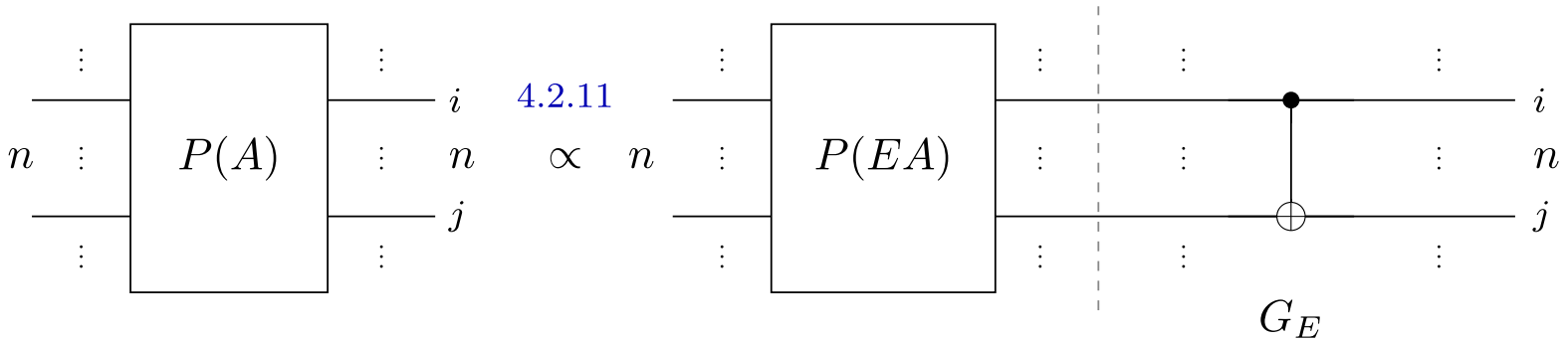
Proposition 4.2.13.

A PNF with invertible biadjacency matrix can be diagrammatically rewritten into a CNOT circuit.

Proof.

1. Reduce A to I by Gaussian elimination.
2. For each row addition E, use Lemma 4.2.11 backwards to extract its CNOT G_E:

$$P(A) = P(E^2A) \rightsquigarrow G_E \circ P(EA).$$



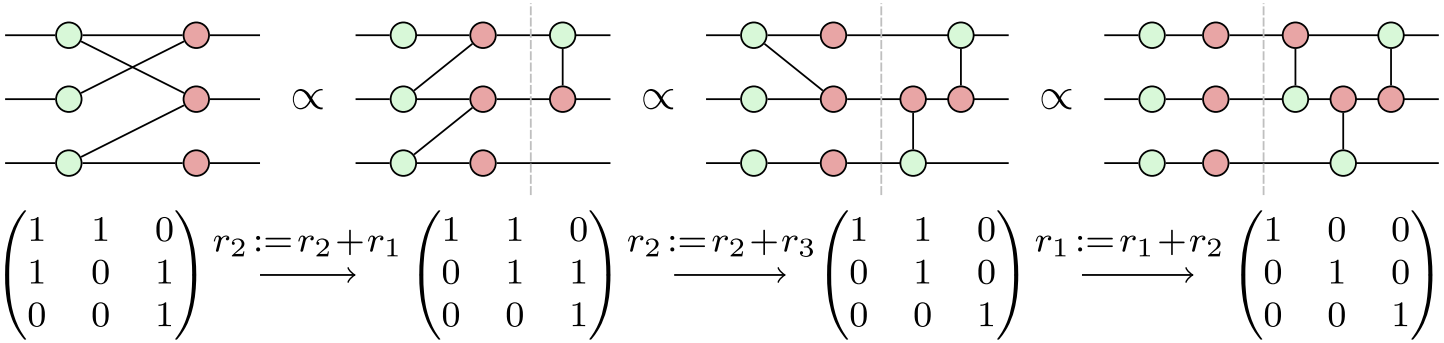
Each new gate is inserted immediately after the remaining PNF.

3. At $P(I)$, remove the two-legged identity spiders. The remaining CNOT circuit executes the row additions in reverse order. The procedure is efficient by Algorithm 1.

CNOT extraction

EX.

The diagram to the left of the dashed line stays in PNF, with the row-reduced matrix:



Unitary phase-free diagrams

Theorem 4.2.14.

Any unitary phase-free ZX diagram can be efficiently rewritten into a CNOT circuit.

Proof.

1. By Theorem 4.2.10 a unitary phase-free diagram can be simplified to PNF.
2. By Lemma *4.5 its biadjacency matrix is invertible.
3. Proposition 4.2.13 extracts a CNOT circuit.

$$D \rightsquigarrow P(A) \rightsquigarrow C, \qquad A \text{ invertible.}$$

■

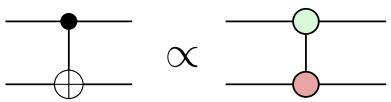
Universality for CNOT

Corollary *5. Up to nonzero scalar.

For every CNOT circuit $C : n \to m$, there is a phase-free diagram $D : n \to m$ with $[\![D]\!] \propto [\![C]\!]$.

Proof.

Replace each CNOT by its phase-free diagram, and keep the same wiring:



Composition and tensor product preserve proportionality.

■

Source

$*$ marks an added or specialised statement. Adapted from Kissinger–van de Wetering, *Picturing Quantum Software*, Chapters 3–4 (CC BY-NC 4.0).

https://zxcalc.github.io/book/html/main_htmlch4.html